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Web sites typically use cookies for the following reasons:

- To collect demographic information about who has visited the Web site.
- Sites often use this information to track how often visitors come to the site and how long they remain on the site.
- It helps to personalize the user's experience on the Web site.
- Cookies can help store personal information about users so that when a usersubsequently returns to the site, a more personalized experience is provided.

If you ever returned to a site and have seen your name mysteriously appear on the screen, it is because on a previous visit, you gave your name to the site and it was stored in a cookie. A good example of this is the way some online shopping sites will make recommendations to users based on their previous purchases. It helps to monitor advertisements. Cookies do not act maliciously on computer system. They are merely text files that can be deleted at any time.

Cookies cannot be used to spread viruses and they cannot access your hard drive. However, any personal information that you provide to a Web site, including credit card information, will most likely be stored in a cookie unless the cookie feature is explicitly turned off in your browser. This is the way in which cookies threaten privacy.

Firewall and Proxy Servers

A firewall is a computer network security based system that monitors and controls incoming and outgoing network traffic based on predefined security rules. A firewall commonly establishes a block between a trusted internal computer network and entrusted computer outside the network. They are generally categorized as network-based or host-based. Network based firewalls are positioned on the gateway computers of LANs [local area Network], WANs [Wide Area Network] and intranets. Host-based firewalls are positioned on the network node itself. The host-based firewall may be a service as a part of the operating system or an agent application such as endpoint security or protection. Each has advantages and disadvantages. However, each has a role in layered security. Firewalls also vary in type depending on where communication originates, where it is intercepted, and the state of communication being traced. **Figure 17.7** shows the working of firewall server.

A proxy server acts as an intermediary between the endusersand a web server. A client connects to the proxy server, requesting some service, such as a file, connection, web page, or other resources available from a different server. The proxy server examines the request, checks authenticity and grants the request based on that. Proxy servers typically keep the frequently visited site addresses in its cache which leads to improved response time.

Figure 17.8 shows the working of a proxy server.